Section A: / *Afdeling A:* 3 **[20]**

Mobile operators focus on retention rather than acquisition since the cost of acquisition is typically five times higher than the cost of retention. The average monthly customer loss ratio (i.e., the percentage of customers that stop using a product in a given month) is approximately 4% for the mobile operator SUNMobile. To reduce the customer loss ratio, SUNMobile has hired a data analytics team to develop a model that can predict whether a customer is likely to leave SUNMobile in the future or not. The steps that the data analytics team proposed to create such a predictive model are illustrated in Figure 1.

Mobiele operateurs fokus op die retensie van kliënte eerder as op die verkryging van kliënte omdat die koste van verkryging tipies vyf keer meer is as die retensie koste. Die gemiddelde maandlikse kliënte verlies verhouding (i.a.w., die persentasie kliënte wat ophou om 'n produk te gebruik tydens die maand), van die mobiele operateur SUBMobile is ongeveer 4%. Om die gemiddelde kliënte verlies verhouding te verminder, het SUNMobile 'n data-ontledingspan aangestel om 'n model te bou wat kan voorspel of 'n kliënt moontlik die maatskappy gaan verlaat in die toekoms of nie. Die stappe wat die data-ontledingspan voorgestel het om die model te ontwikkel word in Figuur 1 uitgebeeld.

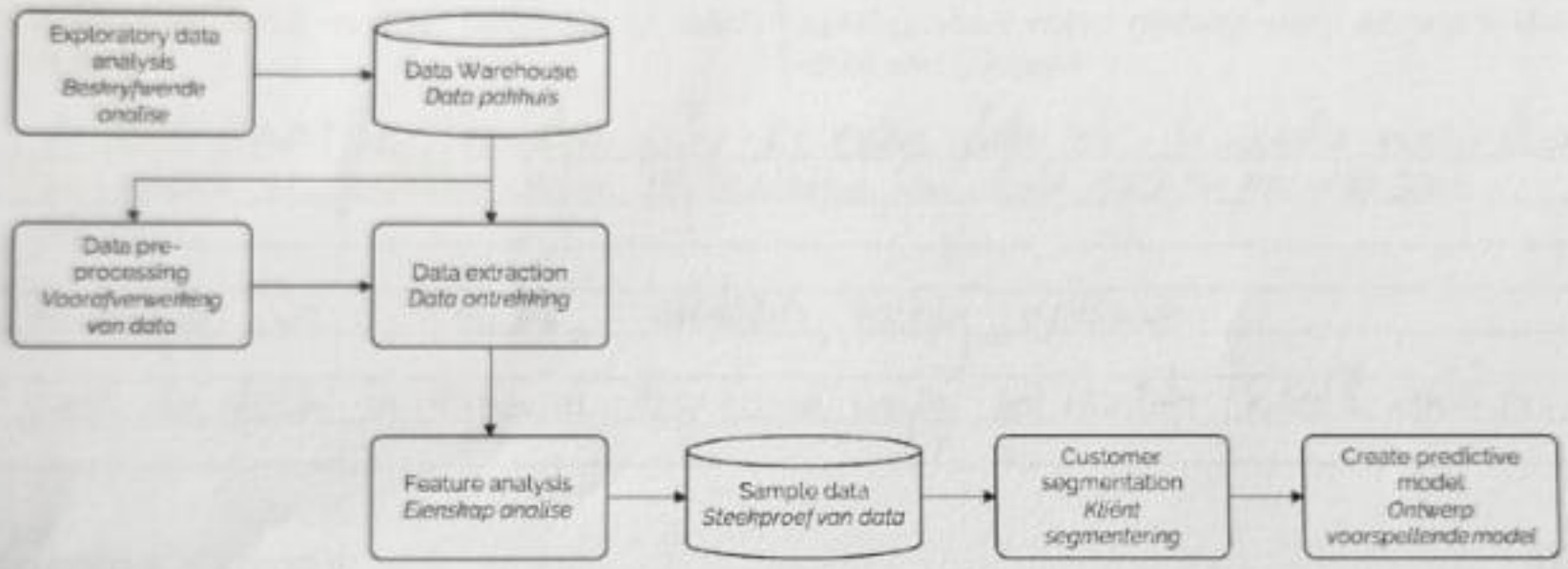


Figure 1: Proposed process to develop a predictive model
Figuur 1: Proses wat voorgestel is om 'n voorspellende model te ontwikkel

Question 1: / Vraag 1: **[3]**

Figure 1 illustrates the data analytic process as a linear process and not a non-linear process. Is this depiction accurate? Support your answer with an example relevant to the case study.

Figuur 1 illustreer die data-analitiese proses as 'n lineêre proses en nie as 'n nie-lineêre proses. Is hierdie uitbeelding akkuraat? Ondersteun jou antwoord met 'n voorbeeld wat relevant is tot die gevallestudie.

This is not accurate. Data analytics are often repetitive and looping processes, some might even call recursive.

An example would be in clustering data → agglomerative clustering needs multiple repetitions before it finds clusters and segments that suit the data.

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4

Question 2: / Vraag 2: **[3]**

Given that structured data is typically stored in a data warehouse, why is a data pre-processing step proposed?

Gegewe dat gestruktureerde data tipies in 'n datapakhuis gestoor word, waarom word 'n data voorafverwerking stap voorgestel?

The data stored in the warehouse is not all in the format that is required for use in analytics - it is stored in one huge table, rather than smaller tables that are needed specifically for this application.

Question 3: / Vraag 3: **[3]**

Assuming that computational processing is unlimited, why is a sample of the data set created? How many samples should be created and what is the purpose of each sample?

Met die veronderstelling dat rekenaar verwerking onbeperk is, hoekom word 'n steekproef van die data geneem? Hoeveel steekproewe moet geskep word en wat is die doel van elkeen?

A model cannot be trained on only the entire data set. it needs new data to validate its predictions before implementation → data that it has not seen before.

the samples are: training, ~~validation~~, testing, validation.

since the data is expensive, they cannot simply get new data to test the model on.

Question 4: / Vraag 4: **[3]**

Is it more appropriate to use hold-out sampling or out-of-time sampling for the provided case study. Motivate your answer.

Is dit meer gepas om uithousteekproefneming of uit-tyd-steekproefneming vir die gegewe gevallestudie te gebruik. Motiveer jou antwoord.

likely out-of-time sampling. this is because the data is not needed immediately, ie like stock markets, and should be done based on old data in concurrence with new/current data.

Question 5: / Vraag 5: **[2]**

What data quality issue will occur based on the information provided in the description of the problem?

Watse datakwalitietkwessie gaan jy te teë kom gebaseer op die inligting verskaf in die beskrywing van die probleem?

obsolescence. longevity. since new ~~out~~ competitors are popping up, circumstances that might have explained previous trends are no longer valid.

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Question 6: / Vraag 6: [2]

The data analytics team advised that client segmentation should be performed to identify different customer loyalty groups before a predictive model is developed. How can the outcome of the data segmentation be used when developing the predictive model? Provide two different options.

Die data-ontledingspan het voorgestel dat kliënt segmentering uitgevoer moet word om verskillende lojaliteitgroepe te identifiseer voordat 'n voorspellende model geskep word. Hoe kan die uitkoms van die kliënt segmentering stap gebruik word in die ontwikkeling van die model? Noem twee moontlikhede.

depending on the segmentation it could benefit in two ways:

- Age, what age demographic is unsubscribing? this might help target products differently.
- Income. what class of person is unsubscribing? perhaps a predictive factor is price increases...

Question 7: / Vraag 7: [4]

To implement customer segmentation the data analytics team used k-means clustering where the value of k was smaller than the number of instances in the data set. Upon analysing the clustering results the team found that one of the clusters contained no instances. Is it possible to have clusters with no instances? Explain your argument with an example.

Om kliënt segmentering te implementeer het die data-ontledingspan die k-gemiddelde groepering metode gebruik met 'n k waarde wat kleiner is as die aantal gevalle in die datastel. Toe die span die resultate analiseer het hulle bevind dat een van die groepe geen gevalle bevat nie. Is dit moontlik om groepe te vind sonder enige gevalle? Verduidelik jou stelling deur gebruik te maak van 'n voorbeeld.

I would say it is not possible. how can there be a group of people that does not include any people?

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Section B: ***/ Afdeling B:*** **[26]**

Intrauterine growth restriction (IUGR) is one of the most common causes of stillbirths (i.e., death or loss of a baby before or during delivery). To help with the early detection of IUGR a model can be developed that predicts whether the fetal weight will be below the 10th percentile at 34 to 37 weeks based on measurements taken at 20 to 24 weeks gestation. To build such a predictive model, two data sets were collected:

- The Week20 data set contains information collected from mothers and their fetuses at 20-24 weeks gestational age. The data set contains 4764 rows, and three features, and is in a one-row-per-patient format.
- The Week34 data set contains the information collected from mothers and their fetuses at 34-37 weeks gestational age. The data set contains 2767 rows, and three features, and is in a one-row-per-patient format.

A description of each feature is provided in Table 1.

Intrauteriene groeibeperking (IUGB) is een van die mees algemene oorsake van stilgeboorte (m.a.w., die dood of verlies van 'n baba voor of tydens geboorte). Om te help om IUGB vroegtydig te identifiseer kan 'n model geskep word wat voorspel of die fetale gewig van 'n fetus onder die 10de persentiel sal wees op 34 tot 37 weke gebaseer op die metings wat ingesamel is tussen 20 en 24 weke van swangerskap. Om 'n voorspellende model te skep, is twee datastelle ingesamel:

- *Die Week20-datastel bevat die inligting van moeders en hul fetusse op 20-24 weke van swangerskap. Die datastel bevat 4764 rye, drie eienskappe en is in 'n een-ry-per-pasiënt formaat.*
- *Die Week34-datastel bevat die inligting van moeders en hul fetusse op 34-37 weke van swangerskap. Die datastel bevat 2767 rye, drie eienskappe en is in 'n een-ry-per-pasiënt formaat.*

'n Beskrywing van elke eienskap word in Table 1 verskaf.

Table 1: Description of features
Tabel 1: Beskrywing van eienskappe

Data set *Datastel*	Feature *Eienskap*	Type *Tipe*	Description *Beskrywing*
Week20	patientID	Numerical *Numeries*	Unique patient identifier *Unieke pasiënt identifiseerder*
Week20	duration	Numerical *Numeries*	The total time (in seconds) that the heart rate of the fetus was recorded *Die totale tyd (in sekondes) wat die hartklop van die fetus opgeneem is*
Week20	beatslost	Numerical *Numeries*	The number of times irregular heartbeats were observed in the recording of the heart rate of the fetus *Die aantal here wat abnormale hartklop waargeneem was tydens die opname van die fetus se hartklop*
Week34	patientID	Numerical *Numeries*	Unique patient identifier *Unieke pasiënt identifiseerder*
Week34	IUGR3	Categorical *Kategories*	Whether or not the IUGR was less than 3% *Of die IUGR minder as 3% was of nie*
Week34	IUGR10	Categorical *Kategories*	Whether or not the IUGR was less than 10% *Of die IUGR minder as 10% was of nie*

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Question 1: / Vraag 1: [2]

To create an analytical base table, the data in the Week20 data set must be combined with the Week34 data set. To combine the data sets you plan to use the left_join function in R. However, when executing your code you received the error: *Could not find function left_join*. What does the error message mean and what steps should you take to resolve the error?

Om 'n analitiese basis tabel te skep moet die Week20-datastel gekombineer word met die Week34-datastel. Jy beplan om die R funksie left_join te gebruik om die datastelle te kombineer. Wanneer jy die kode egter probeer uitvoer ontvang jy die volgende fout boodskap: *Could not find function left_join*. Wat beteken die fout boodskap en watse stappe moet jy neem op die fout op te los?

you used the wrong function beside.

the package for left-join was not installed. you need to find the package to install it and type:

install.package(" ")

Question 2: / Vraag 2: [2]

Assume that you fixed the error in the previous question and that the data sets have been loaded into R as data frames. What will the dimensions of the object returned by the R code *left_join(Week34, Week20)* be? Note any assumptions that you made.

Aanvaar dat jy die fout in die vorige vraag reggestel het en dat die datastelle in R as data frames gelaai is. Wat is die dimensies van die voorwerp wat deur die R kode *left_join(Week34, Week20)* teruggestuur word? Noem enige aannames wat jy maak.

assuming that patient ID is matched up, and that left-join only adds complete rows:

5 x 4764

Question 3: / Vraag 3: [2]

Should the feature IUGR3 be used as a descriptive feature to develop the predictive model? Motivate your answer.

Moet die eienskap IUGR3 ingesluit word as 'n beskrywende eienskap wanneer die voorspellende model ontwikkel word? Motiveer jou antwoord.

No, although it is somewhat redundant, since it is already implied by IUGR10 → which is the actual metric / feature that is predicted.

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Before combining the two data sets, a data quality report was generated for each data set. The data quality report for the Week20 data set is provided in Table 2.

Voordat die twee datastelle gekombineer is, is 'n datakwaliteitverslag vir elkeen van die twee datastelle gegenereer. Die datakwalitietverslag vir die Week20-datastel word in Tabel 2 verskaf.

Table 2: Data quality report for the Week20 data set
Tabel 2: Datakwaliteitverslag vir die Week20-datastel

	Count	%Miss.	Card.	Min	1st Qrt.	Mean	Median	3rd Qrt.	Max	Std Dev.
patientID	4764	0.0	4764	1	1191.8	2382.5	2382.5	3573.3	4764	1375.4
duration	4764	0.0	770	0	1460	1528.6	1558.0	1660.0	3362.0	319.48
beatslost	4764	0.0	41	0	8.00	13.0	12.0	18.0	40.0	7.3

Question 4: / Vraag 4: **[2]**

Consider the data quality report provided in Table 2. Does the table contain any potential missing values? Motivate your answer.

Oorweeg die datakwaliteitsverslag wat in Tabel 2 verskaf is. Bevat die tabel enige potensiële verlore waardes? Motiveer jou antwoord.

Yes. there are other metrics that are usually not present in the table, such as 2nd Quart

Question 5: / Vraag 5: **[3]**

The feature beatslost is correlated with the target feature and with the feature duration. What type of feature is the feature beatslost? Would you recommend removing the feature or not? Motivate your answer.

Die eienskap beatslost korreleer met die teikeneienskap en met die eienskap duration. Watse tipe eienskap is die eienskap beatslost? Sal jy aanbeveel om die eienskap beatslost te verwyder of te behou? Motiveer jou antwoord.

it is a descriptive feature. Perhaps it can be classified as redundant, since it is correlated with more than just the target, but it may be an important metric to use for prediction. Do not remove the feature.

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Question 6: / Vraag 6: **[2]**

With the goal in mind of building an accurate model, you decide to create a derived feature named my_feature. The feature my_feature is the ratio between the feature beatslost and the feature duration. How can you add the feature my_feature to the Week20 data set using dplyr functions? Provide the relevant R code.

Met die doel in gedagte om 'n akkurate model te skep, besluit jy om 'n afgeleide eienskap met die naam my_feature te skep. Die eienskap my_feature verteenwoordig die verhouding tussen die eienskap beatslost en die eienskap duration. Hoe kan jy die eienskap my_feature by die Week20-datastel byvoeg met dplyr funksies? Verskaf die nodige R-kode.

Question 7: / Vraag 7: **[6]**

To build a model that can predict the target feature IUGR10 you decided to implement a rule-based system. However, most of the descriptive features considered in the problem are numerical features. It is not advised to use numeric features in a rule-based system since it is very unlikely that the rule set will cover unseen data instances. The data mining system LERS uses hierarchical clustering to discretise numerical values into clusters before rule induction is applied. Illustrate how hierarchical clustering can be applied to create discretised groups by clustering the data provided in Table 3 into one cluster using the AGNES algorithm, single link and Manhattan distance. Complete Table 4 using the results obtained.

Om 'n model te skep wat die teikeneienskap IUGR10 kan voorspel, het jy besluit om 'n reëlgebaseerde stelsel te implementeer. Die meeste van die beskrywende eienskappe wat oorweeg word, is egter numeriese eienskappe. Dit word nie aanbeveel om numeriese eienskappe in reëlgebaseerde stelsels te gebruik nie aangesien dit baie onwaarskynlik is dat die reëlstel ongesiene data gevalle sal dek. Die dataontginningstelsel LERS gebruik hiërargiese groepering om numeriese waardes in groepe te diskretiseer voordat reëlinduksie toegepas word. Illustreer hoe hiërargiese groepering toegepas kan word om diskrete groepe te skep deur die data wat in Tabel 3 verskaf is te groepeer in een groep deur die AGNES-algoritme, enkel ketting en Manhattan afstand te gebruik. Voltooi Tabel 4 deur die resultate wat verkry is.

Table 3: Numerical data

Tabel 3: Numeriese data

ID *ID*	1	2	3	4	5	6	7	8
Value *Waarde*	0.1	0.12	0.25	0.5	0.63	0.7	0.75	0.9
Target *Teiken*	1	1	1	1	0	0	0	0

Manhattan: $\left(\sum_{i=1}^{n} (a_i - b_i)^1\right)^{1/1}$

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10

Table 4: Results of applying hierarchical clustering

Tabel 4: Resultate van die toepassing van hiërargiese groepering

Iteration *Iterasie*	Cluster assigned per iteration per ID *Groepering toegewys per iterasie per ID*							
	1	2	3	4	5	6	7	8
0	A	B	C	D	E	F	G	H
1	A	A	C	D	E	F	G	H
2	A	A	C	D	E	F	F	H
3	A	A	C	D	F	F	F	H
4	A	A	A	D	F	F	F	H
5	A	A	A	C	F	F	C	H
6	A	A	A	F	F	F	F	F
7	A	A	A	A	A	A	A	A

Question 8: / Vraag 8: **[2]**

The clustering algorithm described in question 7 should be stopped just before the accuracy that can be obtained when using a rule set will decrease as a result of merging clusters. Based on this statement and the result obtained in question 7, how many clusters should you discretise the given values into? Motivate your answer

Die groepering algoritme wat in vraag 7 beskryf was, moet gestop word net voordat die akkuraatheid wat verkry kan word met 'n reëlstel sal afneem wanneer groepe saamgevoeg word. Op grond van hierdie stelling en die resultaat wat in vraag 7 verkry is, in hoeveel groepe moet jy die gegewe waardes diskretiseer? Motiveer jou antwoord.

I believe 4 clusters is best, since at that point, none of the features are clustered with other features with different targets, however 2 clusters is also a good option to avoid over-fitting → also the model is still 87.5% accurate at this point.

Question 9: / Vraag 9: **[2]**

Do you think that discretising numerical values is more likely to result in a model that overfits or underfits the data in the presence of target noise? Assume that the algorithm used to develop the model is PRISM.

Dink jy dat om numeriese data in kategoriese data te verander is meer waarskynlik om te lei tot 'n model wat die data oorpas of onderpas in die teenwoordigheid van teikengeraas? Neem aan dat die model ontwikkel word met die PRISM algoritme.

overfits. if an instance is far enough away from another cluster, it will likely form its own cluster and then result in either inaccurate or overfit models that try to include such case.

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Question 10: / Vraag 10:

[3]

Analyse the data provided in Table 4 using the cut edge weight statistic (CEWS). Is instance four a noisy instance? Motivate your answer with reference to the CEWS algorithm.

Analiseer die data in Tabel 4 deur gebruik te maak van die sny grens gewig statistiek (SGGS). Bevat die teiken van geval vier ruis? Motiveer jou antwoord deur na die SGGS algoritme te verwys.

Yes, it lies between two main clusters, even as they grew. CEWS looks for dozen of data within clusters to data within other clusters, and instance 4 is right in between

↑

please give

bigger lines to write on next time? these are way too small...

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Section C: / *Afdeling C:*

[26]

Company X wants to develop a predictive model for milling machines with the goal of indirectly improving milling machine availability. To develop this predictive model, Company X collected 10 000 instances, each described by the eight features listed in Table 5.

Maatskappy X wil 'n voorspellende model vir freesmasjiene ontwikkel met die doel om indirek die beskibaarheid van die masjiene te verbeter. Om die model te ontwikkel, het Maatskappy X 10 000 gevalle ingesamel wat elk beskryf word deur die agt kenmerke wat in Tabel 5 gelys is.

Table 5: Description of features

Tabel 5: Beskrywing van eienskappe

Feature *Eienskap*	Description *Beskrywing*
Productid	The unique machine identifier *Die unieke masjien identifiseerder*
Type	The type of the machine *Die tipe masjien*
Temperature air	The temperature of the air measured in degrees Celsius *Die tempratuur van lug gemeet in grade Celsius*
Temperature machine	The temperature of the machine measured in degrees Celsius *Die tempratuur van die masjien gemeet in grade Celsius*
Rotational speed	The speed of the machine was measured in Radians per second *Die spoed van die masjien wat in Radiale per sekonde gemeet is*
Torque	The torque of the machine measured in Newton meter *Die wringkrag van die masjien gemeet in Newton meter*
Tool wear	The usage of the current tool measured in minutes *Die gebruik van die hudige instrument gemeet in minute*
Failed	A binary feature indicating whether the milling machine failed (TRUE) or not (FALSE) *'n Binêre kenmerk wat aandui of the freesmajien gevaal het (WAAR) of nie gevaal het (ONWAAR).*

Question 1: / Vraag 1:

[3]

Consider the density plot of the descriptive feature Tool wear. Why is the plot inappropriate to establish if there is a relationship between the feature Tool wear and the target feature Failed?

Oorweeg die digtheidsplot van die beskryfwended kenmerk *Tool wear*. Waarom is die plot onvanpas om vas te stel of daar 'n verband tussen die kenmerk *Tool wear* en die teiken kenmerk *Failed* is?

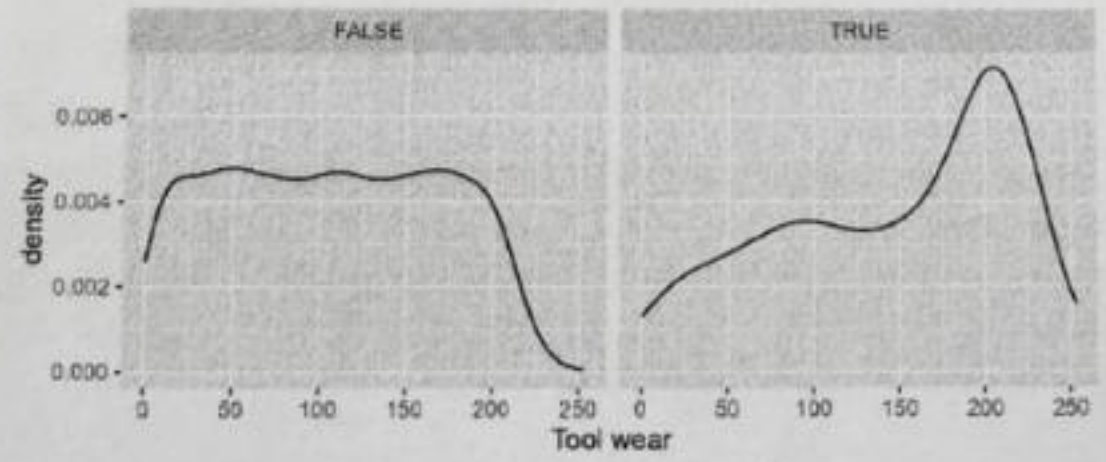


Figure 2: Density plot of Tool wear by Failure mode

Figuur 2: Digtheid plot van Tool wear by Failure tipe

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the graph uses a line that is overfit to the points presented.
there are also significant outliers.

Question 2: / Vraag 2: **[3]**

The tally of instances per Tool wear (binned into five groups) and *Failed* is provided in Table 6. Using, Table 6, provide the rule set that will be learned if the OneR algorithm is applied.

Die telling van gevalle per *Tool wear* (wat in vyf groepe ingedeel was) en *Failed* word in Tabel 6 verskaf. Bepaal die reëlstel wat geleer sal word indien die OneR algoritme toegepas word.

Table 6: Observed frequencies
Tabel 6: Waargeneemde frekwensies

	Failed	
Tool wear	TRUE / *WAAR*	FALSE / ONWAAR
(0, 50.6]	28	1644
(50.6, 101]	42	1583
(101, 152]	34	1552
(152, 202]	51	1602
[202, 253)	69	395

IF tool wear > 202
THEN
Failed = TRUE

Question 3: / Vraag 3: **[3]**

Discuss any potential issues with the rule set obtained in Question 2. How can the training data be modified to obtain an improved rule set using the OneR algorithm? Assume new data cannot be collected.

Bespreek enige potensiële probleme met die reëlstel wat in Vraag 2 verkry is. Hoe kan die opleidingsdata gewysig word om 'n verbeterde reëlstel te kry as die OneR algoritme gebruik word? Aanvaar dat nuwe data nie ingesamel kan word nie.

~~the rule set is too specific → it will not be very applicable to real world scenarios.~~
~~the model predicts wrong - against our better judgement. you would think that more wear → more faults, but not in this limited set of data.~~
other ~~dom~~ descriptive features must be included
it is not well fit enough

Question 4: / Vraag 4: **[3]**

The chi-squared test can be used to assess the association between two categorical features. Populate Table 7 with the expected frequency values between the features Tool wear and Failed.

Om die assosiasie tussen twee kategoriese kenmerke te assesseer kan die chi-kwadraattoets uitgevoer word. Voltooi Tabel 7 deur die verwagte frekwensiewaardes tussen die kenmerke *Tool wear* en *Failed* te bereken.

$\frac{m-j}{m}$

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Table 7: Expected frequencies
Tabel 7: Verwagte frekwensiewaardes

Tool wear	Failed: TRUE / *WAAR*	Failed: FALSE / ONWAAR
(0, 50.6]	16.7%	98.3%
(50.6, 101]	1.39%	97.4%
(101, 152]	2.14%	97.8%
(152, 202]	3.04%	96.9%
(202, 253]	14.9%	85.1%

Question 5: / Vraag 5: **[3]**

How can the expected frequencies be calculated and presented in a table like Table 6 using function from the tidyverse package? Provide the appropriate R code. Assume that you have a data set with the columns Tool wear (binned) and Failed, where each instance represents one measurement.

Hoe kan die verwagte frekwensies bereken en aangebied word in 'n tabel soortgelyk aan Tabel 6 deur gebruik te maak van funksies omvat in die tidyverse pakket? Verskaf die R kode. Aanvaar dat jy 'n datastel het met die kollome *Tool wear (binned)* en *Failed*, waar elke geval een meting verteenwoordig.

Question 6: / Vraag 6: **[1]**

Provided that the chi-square statistic is large and the p-value is 2.16 e-47, the null hypothesis should be _____?

Met die verstande dat die chi-kwadraat statistiek groot is en die p-waarde 2.16 e-47 is, moet die nulhipotese _____ word?

~~the machine fails~~. the machine does not fail.

Question 7: / Vraag 7: **[3]**

When selecting a performance measure to evaluate the predictive model, the company advised you that any potential failures could potentially injure machine operators. To prioritise safety above downtime, which classification metric should you use? Motivate your answer.

Die maatskappy het gevra dat jy in ag moet neem dat enige moontlike masjien foute masjienoperateurs kan beseer. Watter klassifikasiemaatstaf moet jy gebruik om veiligheid bo stilstand te prioritiseer? Motiveer jou antwoord.

Precision. A high precision score results in a large number of ~~Recenion~~ this score depicts how often the model "misses" a fault. It is better to over estimate (ie. predict fault when it isn't), than miss faults.

↓ $\frac{TP}{TP+FN}$ ↑ → Recenion or Precision

	T	F
T	TP	FN
F	FP	

$\frac{TP}{TP+FP}$ ↓ ↑

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Question 8: / Vraag 8:

Consider the receiver operating characteristic (ROC) curve provided in Figure 3. Explain how curve was generated and how the curve can be used to find an appropriate balance between safety and down-time (by referring to Figure 3).

[4]

Beskou die ontvanger-bedryfskenmerk (OBK) kromme wat in Figuur 3 verskaf word. Verduidelik hoe die kurwe gegenereer is en hoe die kurwe gebruik kan word om 'n gepaste balans tussen veiligheid en stilstand te vind, deur te verwys na Figuur 3.

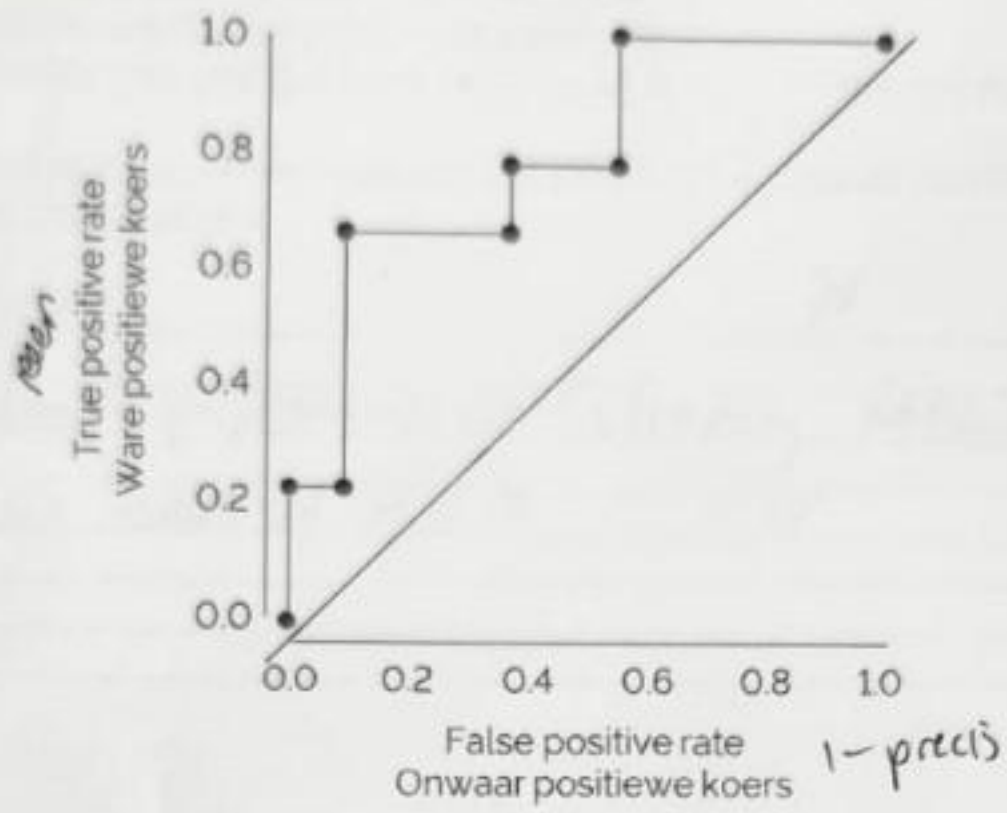


Figure 3: ROC curve
Figuur 3: OBK kromme

The curve is generated by taking precision & recen values at different rates.
this can be used to map the appropiate point on for operations → ie what point is the best of both worlds, or has the best ~~accuracy~~ metric for our use (✓✓)

For this circumstance, a false positive rate of 0.6 means we do not ~~make~~ miss any true positives, optimising 'safety' over downtime. ✓

Question 9: / Vraag 9:

[3] 3

Company X also requested that the speed of the model developed is considered. More specifically an accurate model that uses less features should be preferred. To find an accurate model that use less features Algorithm 1 is proposed. What method of feature selection does Algorithm 1 apply? Are their any methodological mistakes in Algorithm 1?

Maatskappy X het ook versoek dat die spoed van die model wat ontwikkel is, oorweeg word. Meer spesifiek moet 'n akkurate model wat minder eienskappe gebruik, verkies word. Algoritme 1 is voorgestel om 'n akkurate model te vind wat min eienskappe gebruik. Watter metode van eienskappe seleksie pas Algoritme 1 toe? Is daar enige metodologiese foute in Algoritme 1?

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Algorithm 1: Feature selection
Algorithme 1: Eienskap seleksie

1. For every unique combination of features
   Vir elke unieke kombinasie van eienskappe
2. Train a model using the training data set
   Lei 'n model op deur gebruik te maak van die opleidingdatastel
3. Measure the performance of the model on the training data set
   Meet die prestasie van die model op die opleidingsdatastel
4. Identify all models with sufficient performance
   Identifiseer alle modelle met voldoende prestasie
5. Select the model that use the least number of features from the models identified in step 4
   Kies die model wat die minste hoeveelheid eienskappe gebruik van die modelle wat geïdentifiseer was in stap 4

preferential bias.

you don't measure performance on the training data set. this should be done with an unseen set of data.

2

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Section D: ***/ Afdeling D:*** **[18]**

A farmer is interested in developing a model that can predict wheat yield close to the harvest period. Towards this extent the farmer has collected a data set. The features of the data set are described in Table 8 while the statistical properties of the data set is provided in Table 9.

'n Boer stel daarin belang om 'n model te ontwikkel wat koringopbrengs naby aan die oestydperk kan voorspel. Hiervoor het die boer 'n datastel ingesamel. Die kenmerke van die datastel word in Tabel 8 beskryf terwyl die statistiese eienskappe van die datastel in Tabel 9 verskaf word.

Table 8: Description of features
Tabel 8: Beskrywing van eienskappe

Feature *Eienskap*	Type *Tipe*	Description *Beskrywing*
Longitude	Numerical *Numeries*	Longitude of the plot *Lengtegraad van die plot*
Latitude	Numerical *Numeries*	Latitude of the plot *Breedtegraad van die plot*
Year	Numerical *Numeries*	Year that the measurements were taken *Die jaar wat die metings geneem is*
Rainfall	Numerical *Numeries*	Accumulative rainfall received from the planting date until the date of prediction *Akkumulatiewe reënval ontvang vanaf die plantdatum tot die datum van voorspelling*
Soil	Ordinal *Ordinaal*	Soil quality rating, ranging from 1 (poor soil) to 5 (good soil) *Grondkwaliteitgradering, wat wissel van 1 (swak grond) tot 5 (goeie grond)*
NDVI	Numerical *Numeries*	Normalised difference vegetation index derived from an aerial image *Genormaliseerde verskil plantegroei-indeks afgelei van 'n lugbeeld*
GNDVI	Numerical *Numeries*	Green normalised difference vegetation index derived from an aerial image *Groen genormaliseerde verskil plantegroei-indeks afgelei van 'n lugbeeld*
GSAVI	Numerical *Numeries*	Green soil-adjusted vegetation index derived from an aerial image *Groen grond-aangepaste plantegroei-indeks afgelei van 'n lugbeeld*
SIPI2	Numerical *Numeries*	Structure insensitive pigment index 2 derived from an aerial image *Struktuuronsensitiewe pigmentindeks 2 afgelei van 'n lugfoto*
Yield	Numerical *Numeries*	Wheat yield in kg/hectare *Koringopbrengs in kg/hektaar*

Table 9: Data quality report
Tabel 9: Datakwaliteitverslag

	Count	%Miss.	Card.	Min	1st Qrt.	Mean	Median	3rd Qrt.	Max	Std Dev.
Longitude	1183	0	1183	18.89	18.89	18.89	18.89	18.89	18.89	4e-04
Latitude	1183	0	1183	-33.9	-33.9	-33.9	-33.9	-33.9	-33.9	2e-04
Year	1183	0	2	2020	2020	2020.7	2021	2021	2021	0.5
Rainfall	1183	0	2	402	402	425	402	473	473	33.3
Soil	1183	0	5	1	2	3.2	3	4	5	1.3
NDVI	1183	0	905	0.45	0.79	0.82	0.85	0.88	0.92	8e-02
GNDVI	1183	0	899	0.48	0.68	0.71	0.73	0.76	0.81	6e-02
GSAVI	1183	0	968	0.22	0.50	0.55	0.56	0.60	0.70	8e-02
SIPI2	1183	0	464	0.87	0.91	0.92	0.92	0.04	1.09	1e-02
Yield	1183	0	1127	0.4	2297	3120.5	3132.4	3819.6	6356	1161

[2]

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Question 1: / Vraag 1:

Since you do not understand what good crop yield values are, you are unsure how to determine if the model that you plan to develop performs adequately. What performance measure can you use to determine how your model performs in a unitless manner?

Aangesien jy nie verstaan wat goeie oesopbrengswaardes is nie, is jy onseker hoe om te bepaal of die model wat jy beplan om te ontwikkel voldoende presteer. Watter prestasiemaatstaf kan jy gebruik om te bepaal hoe jou model op 'n eenheidlose wyse presteer?

increase or change in yield

Question 2: / Vraag 2: [2]

When training a rule-based model on the data set you found that the rule set tends to exclude rainfall. Why is rainfall excluded, despite its practical importance in crop yield? Motivate your answer using relevant information contained in Table 9.

Wanneer 'n reëlgebaseerde model op die datastel opgelei word, is daar gevind dat die reëlstel geneig is om reënval uit te sluit. Waarom word reënval uitgesluit, ten spyte van die praktiese belangrikheid daarvan in oesopbrengs? Motiveer jou antwoord deur te verwys na relevante inligting in Tabel 9.

the rainfall amount appears to be relatively constant, with a small std. deviation ✓ this results in it not being directly correlated to the increase.

Question 3: / Vraag 3: [3]

Figure 4 shows the relationship between the descriptive feature SIPI2 and the target feature yield. What do you expect the Pearson correlation value to be? Explain how you derived the answer.

Figuur 4 beeld die verband tussen die beskrywende-eienskap SIPI2 en die teiken-eienskap Yield. Wat verwag jy sal die Pearson korrelasie waarde wees? Verduidelik hoe jy die antwoord afgelei het.

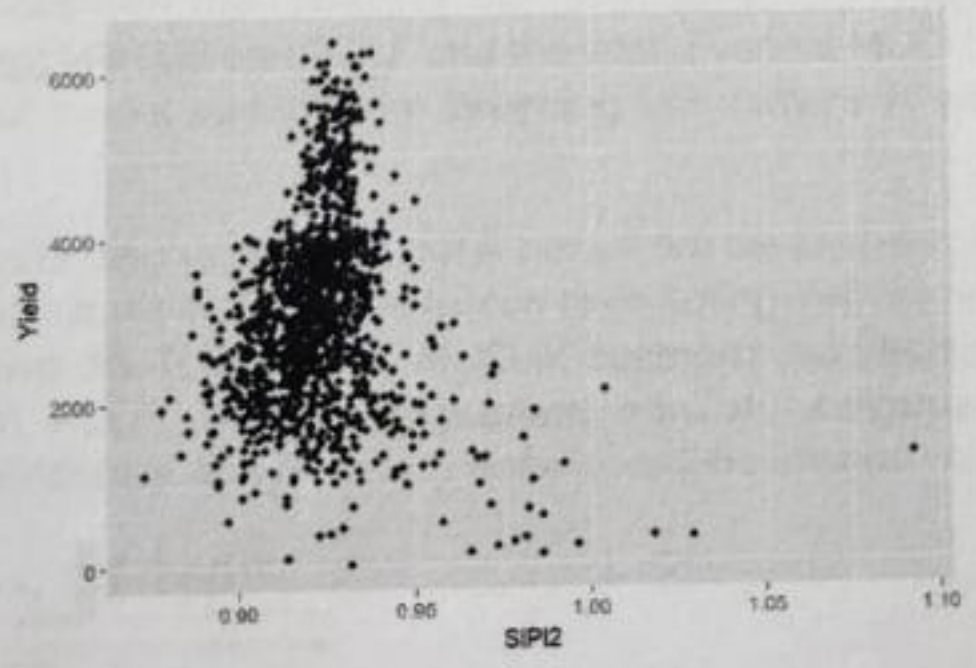


Figure 4: SIPI2 vs Yield
***Figuur 4:* SIPI2 vs Yield**

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0.2 - 0.4. there is little- to- no correlation between these features, and the variety of yield according to SIPI2 is almost non-linear.

Question 4: / Vraag 4: **[3]**

You expect that values for the Soil feature will not always be available. To identify a substitute feature, you decided to calculate and cluster the Pearson correlation between features. Interpret the dendrogram provided in Figure 5 to determine which feature can potentially be used if the Soil feature is unavailable. Motivate your answer by explaining what the provided dendrogram illustrates.

Jy verwag dat waardes vir die Soil eienskap nie altyd beskikbaar gaan wees nie. Om 'n plaasvervangende eienskap te identifiseer, het jy besluit om die Pearson korrelasie tussen kenmerke te bereken en te groepeer. Interpreteer die dendrogram wat in Figuur 5 verskaf word om te bepaal watter kenmerk moontlik gebruik kan word as die Soil eienskap nie beskikbaar is nie? Motiveer jou antwoord deur te verduidelik wat die verskafde dendrogram illustreer.

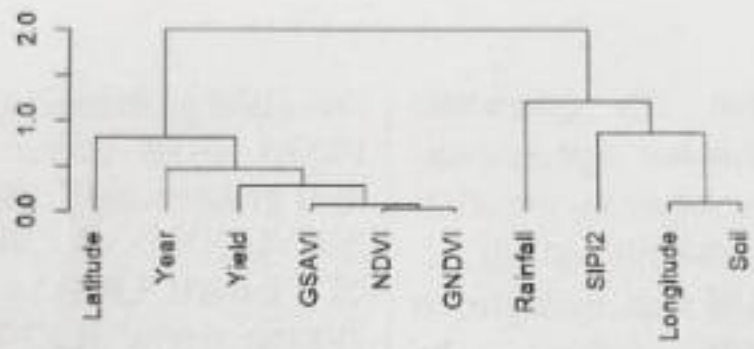


Figure 5: Dendrogram of features
Figuur 5: Dendrogram van eienskappe

this dendogram represents the fact that many of the information features are available in bundles → your location can predict your soil type, and by knowing the year & latitude, you can predict or obtain info on ~~yield~~, yield, etc.

logic

Δquestion To help explain the variance in yield you decided to collect the breeding line of each instance. A wheat breeding line usually indicates the cross made between two wheat variants or parents. For example, the wheat variant with the breeding line "MUTUS/ROLF//MUCUY" consist of a cross between the wheat variant MUTUS and ROLF and the wheat variant MUCUY. Since there are many possible combinations of wheat variants the breeding line feature is expected to have a high cardinality.

Om die verskil in opbrengs te help verduidelik, het jy besluit om die teellyn van elke geval te versamel. 'n Koringteellyn dui gewoonlik die kruising wat tussen twee koringvariante of ouers gemaak was. Die koringvariant met die teellyn "MUTUS/ROLF//MUCUY" bestaan byvoorbeeld uit 'n kruising tussen die koringvariant MUTUS en ROLF en die koringvariant MUCUY. Aangesien daar baie moontlike kombinasies van koringvariante is, kan jy verwag dat die teellyn eienskap 'n hoë kardinaliteit sal hê.

John #1 Apple
Seed

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Question 5: / Vraag 5: **[5]**

How can we cluster the breeding lines into meaningful groups? Discuss how you can measure the similarity between two breeding lines. Assume hierarchical clustering will be applied. Hint: The breeding line MUTUS/ROLF//MUCUY is closer to the breeding line KUTZ//KFA/ROLF than it is to the breeding line KENYA/KACHU//KIDEA since it includes one of the wheat variants.

Hoe kan die teellyne in sinvolle groepe groepeer word? Bespreek hoe jy kan meet hoe soortgelyk twee teellyne is. Veronderstel hiërargiese groepering sal toegepas word. Wenk: Die teellyn MUTUS/ROLF//MUCUY is nader aan die teellyn KUTZ//KFA/ROLF as aan die teellyn KENYA/KACHU//KIDEA aangesien dit een van die koringvariante insluit.

similarity between groups is naturally related to the breeds that are involved in the cross breed pairs.

a heirarchy must be established according to how many breeds are involved → len = more pairs = better relation to another.

thus must be clustered using tags, according to breeds and then a similarity measure must take place.

Question 6: / Vraag 6: **[3]**

Consider the fact that some breeding lines will include multiple variants while other might only consist of two variants. This means that the breeding line MUTUS/ROLF//MUCUY will be as similar to KUTZ//KFA/ ROLF then it is to KENYA/KACHU//KIDEA/ROLF when simply matching variant names. Suggest a way to calculate the similarity measure to account for the number of variants? In other words your measure should mathematically express that MUTUS/ROLF//MUCUY is more similar to KUTZ//KFA/ROLF than it is to KENYA/KACHU//KIDEA/ROLF

Oorweeg die feit dat sommige teellyne veelvuldige variante sal insluit, terwyl ander dalk net uit twee variante bestaan. Dit beteken dat die teellyn MUTUS/ROLF//MUCUY net so soortgelyk aan KUTZ//KFA/ ROLF sal wees as aan KENIA/KACHU//KIDEA/ROLF wanneer ooreenstem met variantname gebruik word. Stel 'n manier voor om die ooreenkomsmaatstaf te bereken om rekening te hou van die aantal variante? Met ander woorde, jou maatstaaf moet wiskundig uitdruk dat MUTUS/ROLF//MUCUY meer soortgelyk is aan KUTZ//KFA/ROLF as aan KENIA/KACHU//KIDEA/ROLF

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7

Section E: / *Afdeling E:* **[10]**

A small data set was collected to recognise human activity from a wearable accelerometer sensor. Table 7 contains instances with the average acceleration in the X, Y and Z directions within a short time window. There is no target feature, instead clustering will be used to determine if there are meaningful groups.

'n Klein datastel is saamgestel om die aktiwiteite wat 'n mens uitvoer deur 'n draagbare versnellingsmeter sensor te herken. Tabel 7 bevat gevalle met die gemiddelde versnelling in die X-, Y- en Z-rigtings binne 'n kort tyd venster. Daar is geen teikeneieskap nie, groepering moet gebruik word om betekenisvolle groepe te identifiseer.

Table 7: Activity data set
Tabel 7: Aktiwiteitdatastel

ID	X	Y	Z
1	-0.154	0.376	0.099
2	-0.103	0.476	-0.027
3	0.228	0.036	-0.251
4	0.330	0.013	-0.263
5	-0.114	0.482	0.014
6	0.295	0.084	-0.297
7	0.262	0.042	-0.304
8	-0.051	0.416	-0.306

Question 1: / Vraag 1: **[7]**

The K-means clustering approach should be applied to the data set provided in Table 8, with k = 2 and using Euclidean distance. Assume that the initial cluster centroids for the clusters C1 and C2 are $\mathbf{c}_1$ = <-0.235, 0.253, 0.438> and $\mathbf{c}_2$ = <0.232, 0.325, -0.159>. Calculate the distance between each data instance and each centroid. Use the results of the calculations to complete Table 8. Round your answers to three decimal figures.

Die K-gemiddelde groepering metode moet toegepas word op die datastel verskaf in Tabel 8, met k = 2 en deur gebruik te maak van Euklidiese afstand. Aanvaar dat die aanvanklike sentroïede van die groepe C1 en C2, $\mathbf{c}_1$ = <-0.235, 0.253, 0.438> en $\mathbf{c}_2$ = <0.232, 0.325, -0.159> is. Bereken die afstand tussen elke data geval en elke sentroïed. Gebruik die resultate van die berekeninge om Tabel 8 te voltooi. Rond jou antwoord af tot drie desimale syfers.

Table 8: Activity data set – distance calculations
Tabel 8: Aktiwiteitdatastel – afstand berekeninge

ID	X	Y	Z	dist($\mathbf{d}_i$, $\mathbf{c}_1$)	dist($\mathbf{d}_i$, $\mathbf{c}_2$)
1	-0.154	0.376	0.099	0.370	0.467
2	-0.103	0.476	-0.027	0.532	0.390
3	0.228	0.036	-0.251	0.858	0.303
4	0.330	0.013	-0.263	0.952	0.343
5	-0.114	0.482	0.014	0.497	0.417
6	0.295	0.084	-0.297	0.922	0.285
7	0.262	0.042	-0.304	0.918	0.319
8	-0.051	0.416	-0.306	0.784	0.332

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Question 2: / Vraag 2: [3]

Calculate the centroid values of cluster 1 after the 1st iteration of the K-means algorithm. Show all your calculations.

Bereken die sentroïed waardes vir groep 1 na die eerste iterasie van die K-gemiddelde algoritme. Wys al jou berekeninge.

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